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نقل الكهرباء
National Grid SA

# TES-P-119.42

## Revision No. 01

## *STATIC VAR COMPENSATOR (SVC)*


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# TRANSMISSION ENGINEERING STANDARD

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## 1.0 SCOPE

This Engineering Standard establishes requirements for Static VAR Compensator (SVC) in Transmission Substations of National Grid Saudi Arabia.

## 2.0 INTRODUCTION

The Static Var Compensator (SVC) is a shunt device of the Flexible AC Transmission Systems (FACTS) family using power electronics to control power flow and improve transient stability on power grids. The SVC regulates voltage at its terminals by controlling the amount of reactive power injected into or absorbed from the power system. When system voltage is low, the SVC generates reactive power (SVC capacitive). When system voltage is high, it absorbs reactive power (SVC inductive). The variation of reactive power is performed by switching three-phase capacitor banks and inductor banks connected on the secondary side of a coupling transformer. Each capacitor bank is switched on and off by thyristor switches (Thyristor Switched Capacitor or TSC). Reactors are either switched on-off (Thyristor Switched Reactor or TSR) or phase-controlled (Thyristor Controlled Reactor or TCR).

## 3.0 SVC V-I CHARACTERISTICS

SVC can be operated in two different modes. In the voltage regulation mode and in var control mode. When SVC is operated in voltage regulation mode, it implements the V-I characteristics as shown in figure 42-01. As long as the SVC susceptance B stays within the maximum and minimum susceptance values imposed by the total reactive power of the capacitor banks (Bcmax) and reactor banks (Blmax), the voltage is regulated at reference voltage Vref. However a voltage droop is normally used (usually between 1% and 5% at maximum reactive power output). The V-I characteristics are described by the following three equations:

SVC in regulation range ($-B_{cmax} < B < B_{lmax}$)

$$V = \frac{I}{B_{c\,max}} \quad \text{(SVC is fully capcitive } B=B_{cmax}\text{)} \qquad \text{(Eq. 42-01)}$$

$$V = V_{ref} + X_S \cdot I$$

$$V = \frac{I}{B_{l\,max}} \quad \text{(SVC is fully inductive } B=B_{lmax}\text{)} \qquad \text{(Eq. 42-02)}$$

Where

V = Positive sequence voltage (p.u)
I = Reactive current
$X_S$ = Slope or droop reactance
$B_{cmax}$ = Maximum capacitive susceptance
$B_{lmax}$ = Maximum inductive susceptance

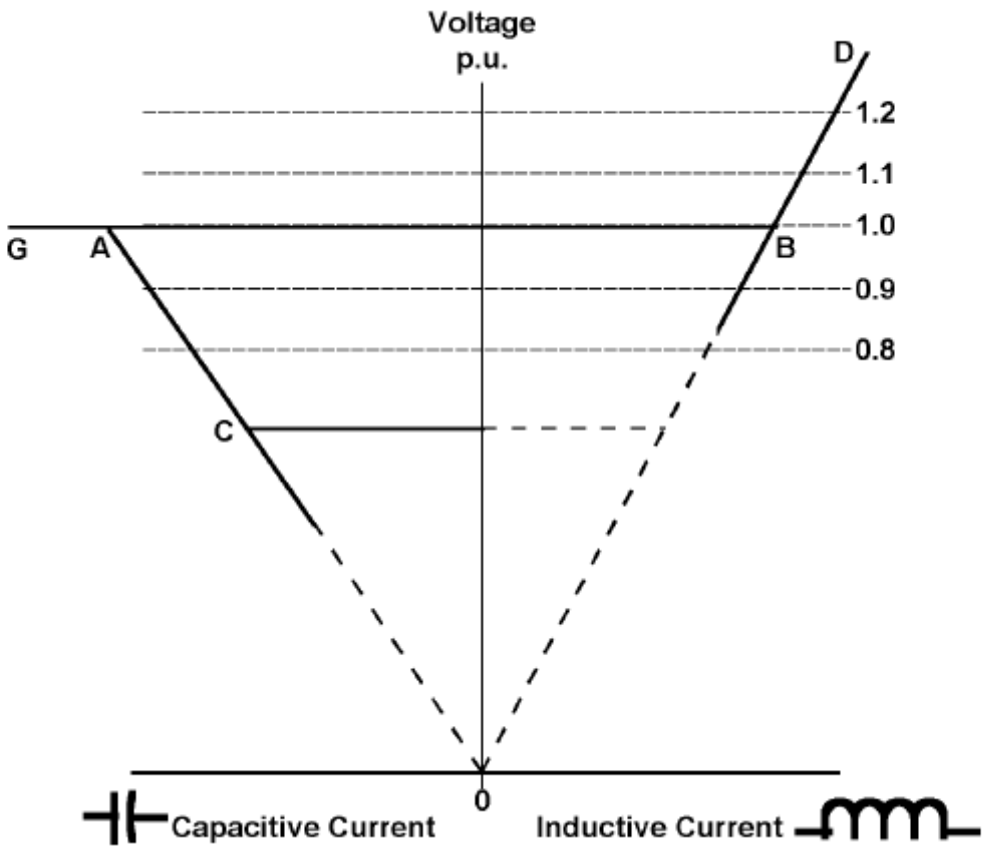



Figure 42-01: SVC V-I CHARACTERISTICS CURVE

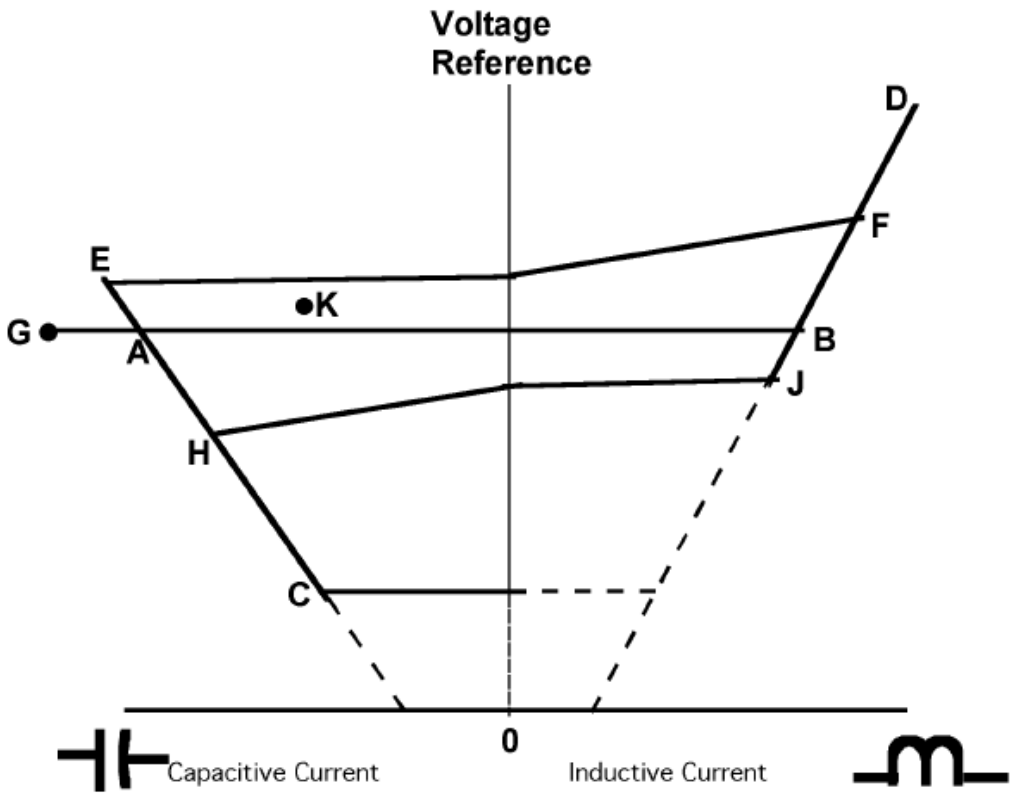
## 4.0 SVC RATING

SVC rating shall be per the following (Refer Figure 42-01 and Figure 42-02). Space indicated as_____below shall be filled up with required data as required for the project during the preparation of Scope of Work and Technical Specification (SOW/TS):

4.1 The SVC shall regulate_____kV bus voltage to a reference voltage of__kV (1.0 per unit) continuously adjustable between ____ per unit and _____ per unit.

4.2 The nominal capacitive reactive power output of the SVC at point A shall be __ per unit on 100 MVA base at 1.0 p.u AC bus voltage and nominal system frequency and ambient temperature per latest revision of 01-TMSS-01.

4.3 The nominal inductive reactive power output of the SVC at point B shall be at__ per unit on 100 MVA base at 1.0 p.u AC bus voltage and nominal system frequency and ambient temperature per latest revision of 01-TMSS-01.




Note: H to J give minimum voltage settings for continuous operation at maximum and minimum slope respectively.
K is the most severe operating condition chosen to define filter component rating

Figure 42-02: V-I CHARACTERISTICS – MAGNIFIED SCALE

4.4 The SVC shall continue to generate maximum capacitive reactive power during a temporary undervoltage down to the value of ____ p.u for the duration of ___ seconds as specified in SOW/TS (Refer point C in Figure 42-01). The SVC (TSCs/TSRs/TCR) may be blocked if the undervoltage persists more than ____seconds.

4.5 The SVC shall continue to absorb reactive power during a temporary overvoltage in a controlled manner up to a value ___ p.u for the duration of ___ seconds and up to the value of ___p.u for the duration ___ second as specified in the SOW/TS (Refer point D in Figure 42-01). This is the minimum requirement and maximum over load and overvoltage requirement based on the network data provided in the SOW/TS shall be determined by the Bidder/Contractor. SVC shall be rated and designed to withstand these overvoltage's.

4.6 The maximum continuous capacitive reactive power output of the SVC is denoted by point E in Figure 42-02 and is obtained when the SVC is operated with minimum droop and maximum reference voltage.

4.7 The maximum continuous inductive reactive power output of the SVC is denoted by point F in Figure 42-02 and is obtained when SVC is operated with maximum droop and maximum reference voltage.

4.8 The reactive power range required is the minimum requirement at the SVC HV bus after allowing for combined effect of all component and control tolerances including that of the transformer. These tolerances shall be stated by the SVC manufacturer and shall be guaranteed and shall be per IEC/IEEE standard by which the equipment is offered.

4.9 The nominal slope of the characteristics shall be adjustable in steps of not greater than ___% between___% and ___% as specified in SOW/TS on a basis of SVC nominal capacitive power rating. (Refer Figure 42-02).

4.10 The SVC shall be capable of continuous operation over the voltage range as indicated in the V-I characteristics.

4.11 (Optional) The short time capacitive reactive power output of the SVC shall be _____ MVAr at ____per unit AC bus voltage for seconds_____(Refer point G in Figure 42-02), following 1 hour operation at or below the level specified in clause 4.2.

4.12 (Optional) Continued controllable conduction of the SVC to absorb reactive power shall be possible up to____ per unit AC bus voltage for ____ seconds following 1 hour operation at or the level specified in clause 4.3.

4.13 SVC power transformer and all bus equipment such as filter branches, thyristor- controlled reactor (TCR) branches, thyristor-switched reactor (TSR), thyristor-switched capacitor (TSC) branches, capacitor bank branches and reactor bank branches (whether at EHV, HV or low voltage (MV) shall be rated to withstand the specified continuous and short term operation and to withstand or to be protected against voltage and current stresses that exceed these conditions. All components of SVC such as filter and capacitor bank branches shall not trip during overload period.

4.14 With SVC Power transformer connected all SVC equipment shall sustain without damage any internal fault in the SVC MV side no matter what fault level is available on the HV system(s). The load break switches alternatively circuit breakers and disconnect switches for the filter branches, disconnector for the Thyristor Controlled Reactor (TCR), Thyristor switched reactor (TSR), and Thyristor Switched Capacitor (TSC) etc., and the circuit breaker for auxiliary transformer/grounding transformer shall be able to withstand the maximum through fault current until the SVC HV breaker or the associated upstream HV breakers has cleared the fault. The fault level at the grounding and auxiliary power transformer shall be limited to 25kA.

4.15 SVC reactive current output shall not be limited while within the specified rating range setting and operating modes. The SVC controller shall be capable

of controlling (switching and blocking) the TCRs/TSRs, TSCs etc., and switching external devices such as bus reactors and capacitors at any time during the transient disturbance.

4.16 Component and Control Tolerance

SVC shall not increase the voltage unbalance at the SVC HV terminals by more than 0.1% due to component, measurement, control system and thyristor firing unbalance tolerances when the voltage balancing control is out of operation or not required.

4.17 Reactive Power Rating Tolerance

The reactive power range required is the minimum requirement after allowing for the combined effect of all component and control tolerance including that of the transformer. These tolerances shall be guaranteed by SVC manufacturer.

4.18 Overload Behavior

Shall be per V-I characteristic as specified in the SOW/TS.

4.18.1 Capacitive and Inductive Overload Cycles

The components of SVC shall be designed for operation consistent with the V-I characteristics defined in the presence of power frequency surges and dips at the SVC HV bus corresponding to the overload cycles specified. The SVC shall not be tripped or prevented from operation during these network conditions. The SVC is allowed to trip for U>1.5 p.u. for more than one cycle.

4.18.2 Capacitive Overload Duty Cycle

Overload duty cycle shall be included in the basic continuous rating. The SVC and in particular SVC power transformer overloading capability shall be designed with the requirement that the transformer operation mode ONAN etc.,) shall not limit the above capability at any time. The SVC must be able to sustain the specified overload cycles as a result of three phase fault or single phase fault (or any combination thereof) on SEC EHV, HV and MV system for the fault duration **(**minimum to maximum) as defined in SOW/TS followed by voltage recovery of 1.3p.u for 3 seconds or 1.5p.u for 1 second or delayed voltage recovery, regardless of the amplitude of the DC component remnant in the SVC thyristor valves.

4.19 Overload Characteristics

Overload capabilities of all plant and material shall be described by the contractor/manufacturer. Overload capabilities of SVC and all the components after factory test shall be submitted**.**

4.20 Undervoltage V-I Characterstics

Unless otherwise specified in the SOW/TS undervoltage characteristics shall be as follows:

4.20.1 The TCR/TSR and TSC branches must be capable of remaining in service continuous down to a voltage of 0.3 p.u. Below this voltage TCR/TSR and TSC shall be blocked and the SVC shall remain connected to the system, ready for immediate voltage control as soon as the HV voltage is above 0.3 p.u.

4.20.2 The SVC and in particular SVC cooling **(**valves and transformer**)** and auxiliary system must be able to sustain the following undervoltage duty cycle:

a. Continuous operation down to 0.9 p.u. HV bus voltage.

b. Short time operation between 0.8 p.u. and 0.9 p.u. HV bus voltage for 30 seconds.

c. Short time operation between 0 and 0.8 p.u. HV bus voltage for 10 seconds.

4.21 Undervoltage Control Strategy

The purpose of the undervoltage control strategy is to prevent an overvoltage condition upon fault clearance as a result of excess amount of MVAr being injected in to the network by the SVC. It shall be noted that the normal SVC operation at undervoltages is to support and balance the voltage in an optimal manner defined by system studies.

4.21.1 An undervoltage condition is detected when the positive phase sequence voltage at the SVC HV bus is below an adjustable lower threshold of 0.30 to 0.95 p.u. (Step size of 0.01 p.u) for an adjustable time of 0 to 5 seconds (step size of 1 ms). This condition is cleared when the positive phase sequence voltage rises above an adjustable upper threshold of 0.30 to 0.95 p.u (step size of 0.01 p.u) for an adjustable time of 0 to 200ms (step size of 1 ms).

4.21.2 Upon detection of an undervoltage condition one of the following control strategies, shall be run. The control strategy to use shall be set at the HMI:

The TCR/TSR and TSC branches must remain in service (deblocked or blocked) during the undervoltage condition, unless it drops below the operational limits.

a. <u>Strategy #1</u>

No special action is required. The SVC reacts to the output of rapid voltage SVC controller.

b. Strategy#2

The output of the SVC controller is set to a preset value (valueB1). This value is adjustable between maximum inductive to maximum capacitive.

c. Strategy#3

All TCR/TSR and TSC branches block. This action occurs at the voltage level where the synchronization for firing pulses is no longer guaranteed.

At voltage recovery, the SVC voltage regulator operation starts from a preset output value B2 (defaulted to 0). The output of the SVC controller is a from this time the normal function of the voltage error.

d. Strategy #4

Contractor/Manufacturer shall develop a control strategy to ensure the overshoot upon fault clearance does not exceed 1.3 p.u at the SVC HV busbar for a maximum duration of 2 cycles of fundamental frequency. Following this, the SVC HV busbar shall be within the voltage limits as specified in 01-TMSS-01. As an option this strategy can take into consideration the loading of the network.

e. Strategy#5

Contractor/Manufacturer shall implement additional undervoltage strategies based on results from the dynamic studies (PSCAD and RTDS). Criteria for detection of undervoltage conditions shall be defined by Contractor/ Manufacturer.

4.21.3 Upon clearing of the undervoltage condition, the rapid voltage SVC controller resumes normal operation but starts from the condition impose during the undervoltage conditions. The SVC controller shall allow enabling or disabling of this function from local, including status indication.

4.22 Overvoltage Control Strategy

4.22.1 The SVC shall remain in operation during temporary overvoltages and reduce them.

4.22.2 The SVC shall not make use of a protection that measures the HV Voltage directly or indirectly to trip the SVC.

4.22.3 An overvoltage condition is detected when the positive sequence voltage measurement as used by the rapid voltage SVC controller, exceeds an adjustable upper threshold of 1.00 to 1.50 p.u (step size of 0.01 p.u) for an adjustable time of 0.000 to 2.000 seconds (step size of 1ms). This condition is cleared when the voltage is below an

adjustable lower threshold of 1.0 to 1.5 p.u. (step size of 0.01 p.u) for adjustable time of 0 to 200 ms (step size of 1ms).

4.22.4 Upon detection of an overvoltage condition, one of the strategies described below shall be run. The control strategy to use shall be set at the HMI:

a. Strategy #1

No special action is required. The SVC reacts to the output of the rapid voltage SVC controller.

b. Strategy #2

Upon detection of an overvoltage condition, full inductive operation of the TCR/TSRs applies. All TSCs shall be blocked.

c. Strategy #3

The Contractor/Manufacturer shall implement additional overvoltage strategies based on results from the dynamic studies (PSCAD and RTDS). Criteria for detection of overvoltage conditions shall be defined by Contractor/Manufacturer in accordance with the relevant SEC/National Grid SA standards.

4.22.5 Under Frequency

In the event that the network frequency drops below the minimum value stated in SOW/TS, the SVC shall not trip and lock out but block all thyristor valves will remain on standby. As soon as the frequency recover to the normal value, the SVC shall go back into normal voltage control mode.

## 5.0 SVC CONTROL OBJECTIVES

The desired function(s) and the priority in which the SVC shall respond to them are:

5.1 Control of three-phase average of positive sequence of the fundamental voltage in steady state, during and after faults, over the full slope range. The SVC voltage controller shall be optimized to rapidly control the HV voltage during and after single phase, phase to phase and three phase faults on the SEC transmission system and the SVC shall not cause overvoltage on the healthy phases. The rapid control function is required to prevent induction motor stalling following single phase to ground faults and associated delayed voltage recovery phenomena. The SVC shall further reduce impact of phase to phase faults. The aim is to maximum voltage support for induction motor loads, without violating voltage limits on line to line voltages or line to ground voltages in the connection point for the SVC during and subsequent to the grid short circuits. SVC shall be designed for unsymmetrical operation for a range specified in SOW/TS for during faults and directly post fault recovery. Unbalanced operation of TCR/TSR/TSC is required.

5.2 Balancing of asymmetries in the system voltage. In steady state, the negative

phase sequence voltage in the grid may reach to two percent (2%). The SVC shall fully balance the negative phase sequence voltage (if sufficient SVC power is available), or bring it below to a preset value (dead-band set from HMI), at its connection to the grid.

The voltage balancing shall be in closed loop mode to reach maximum accuracy.

5.3 The SVC shall automatically control the switching of external shunt capacitor banks or reactors which may be connected to 380kV or 230kV level in the future. This function shall be selectable.

5.4 The SVC shall have the capability to damp all types of power system oscillation. If there is any.

5.5 The SVC shall not trip during the undervoltage conditions as defined in clause 4.21 above. The SVC controller shall:

5.5.1 Prevent harmonic instability due to network resonances.

5.5.2 Not limit the dynamic performance of the SVC.

5.5.3 Not limit the voltage and reactive power control range.

5.6 The SVC shall not trip during the overvoltage conditions as mentioned in clause 4.22 and as specified in the SOW/TS.

## 6.0 SVC RESPONSE & STABILITY

### 6.1 Definitions

Definition of response time, maximum overshoot time and settling time is per clause 3 and 8 of IEEE 1031.

6.2 The change of measured system voltage shall reach 90% of the desired total change within 40ms of the initial control signal. The maximum overshoot shall not exceed 10% of the final change and settling time shall not exceed 100ms, after which the voltage shall be within 5% of the final value. The response is required when the system three-phase fault MVA is between the minimum and maximum value defined in the SOW/TS. This response time shall be achieved without any undervoltage control strategy being implemented.

6.3 The step response requirements shall be met also when switching of a TSC is a part of the step (irrespective of initially discharged or charged TSC). Additionally, the following requirements shall be complied with;

The voltage oscillation at SVC HV side when energizing an uncharged TSC, shall be less than 2.5% in amplitude (positive phase sequence voltage based on HV nominal value), and have a damping better than 33% for each subsequent half cycle). This additional requirement shall be complied with in the Thévenin equivalent used for step response studies and in the "real" network.

6.4 The response of the SVC controller shall be valid over the total SVC Range. This shall be demonstrated by the manufacturer during bid stage, base design stage and during factory performance testing including Real Time Digital Simulator (RTDS) testing.

6.5 The rapid voltage SVC controller shall be designed to ensure stable dynamic behavior of the SVC over the entire short circuit range and system network impedance specified. Gain reduction (if applicable) for voltage controller at low fault level shall be automatically. The step response requirements shall be achieved from minimum short circuit power to maximum short circuit power without any adjustment of the regulator settings. Regulator settings can be optimized if fast (instantaneous) continuous monitoring of short circuit power is provided. It is not allowed to fine tune control parameters manually for each fault level in order to fulfill step response requirements.

6.6 Dynamic performance under very weak condition

The network is considered very weak when the three-phase short-circuit current at the substation is very low (below the specified minimum values) as occurs upon network restoration, for instance. The SVC control system shall ensure stable behavior of the SVC under these conditions (gain supervision and stability control).

6.6.1 Gain Supervision

a. The input signal of the supervisor is the output of the rapid voltage regulator. This signal is filtered so as to retain only the oscillating frequency band specific to instabilities of the compensator voltage regulating loop.

b. Instability is detected when the amplitude of oscillations on the filtered input signal overshoots an adjustable threshold for an adjustable number of times. The response speed of the compensator is gradually reduced at each new overshoot until the amplitude is within acceptable limits.

c. The values of the voltage regulator parameters governing the response speed of the compensator remain at the new values dictated by the supervisor until an operator from Power Control Centre (PCC) commands the return to the initial values or until action by the optimizer as described herein.

6.6.2 Optimization

a. The optimizer re-adjusts the rapid voltage regulator parameters altered by the action of the supervisor so as to optimize the dynamic performance of the compensator relative to the initial settings.

b. The optimizer evaluates the dynamic performances of the static compensator by provoking a small disturbance and then analysing the output of the rapid voltage regulator. It gradually readjusts the parameters until the rapid voltage regulator satisfies its performance criterion or the initial values of the parameters are obtained.

c. Optimization must take account of all the static compensators at the substation or other SVCs electrically close nearby.

Each compensator shall have its own optimizer circuit.

d. The optimizer shall take account of all static compensators on the network. To achieve this, gain freeze send and receive signals shall be provided in the controller.

- The small disturbances generated by the optimizers at various substations must be time-coordinated to ensure balance of the values of the rapid voltage regulator parameters for all the static compensators on the network.

- There shall be a certain delay between the action of the supervisor and activation of the optimizer so as to allow permanent network recovery.

e. When operating near the inductive limit gain optimizer shall perform a test in capacitive direction and when operating near the capacitive limit perform the test in the inductive direction.

6.6.3 The design of these automatic devices shall satisfy the requirements of flexibility for purposes of field adjustment.

6.6.4 These automatic devices shall allow for independent activation or deactivation by Power Control Centre (PCC). The relative value of the rapid voltage regulator parameters affecting these devices shall be displayed and visible to PCC who shall be able to command the return to the initial values.

6.6.5 During bid stage manufacturer shall submit analysis of the behavior of the automatic rapid voltage regulation supervisor/optimizer during disturbances and / or sudden oscillations in the short-circuit level and /or changes in the number of compensators in service. The operating principle of the automatic devices shall be explained and transfer and logic functions shall be provided.

## 7.0 DEGRADED OPERATING MODES

7.1 The SVC controller shall be designed to ensure operation of the SVC following the loss of a TSC or TCR/TSR branches, filter and de-tuned capacitor banks. The controlled reactive power range is reduced to the available reactive power. The range of the available TCR shall not be restricted.

In case of outage of all continuously controllable branches i.e., TCRs the SVC shall remain in operation with the stepwise controlled branches i.e., TSR and TSCs. The control system shall be capable of handling operation with only step wise controlled branches. No hunting shall occur in the span from maximum to low short circuit power. For short circuit power from maximum to minimum hunting shall be prevented without changes in control settings. In the span from minimum to prevailing short circuit power special adaptive control functions are allowed. The short circuit power shall be as defined in SOW/TS.

7.2 The SVC controller shall be designed to ensure operation of the SVC with any

combination of branches in service. The operating mode and performances of the SVC under network disturbances are to be described by the manufacturer. The SVC shall respond as rapidly as possible for maintaining system stability.

7.3 The disconnectors for each of the TSCs and TCR/TSR as well as the load break switches alternatively MV circuit breakers and disconnect switches for the filter branch and de-tuned capacitor banks shall be integrated into the control, monitoring and protection system hardware and software so as:

7.3.1 To correctly identify a failed component within the TSC, TCR/TSR or filter zone that can be disconnected and isolated by the load break switches/disconnector alternatively MV circuit breaker and disconnect switches safely and without causing further damage.

7.3.2 After a failure has occurred such as reactor failures, busbar faults, etc., requiring immediate and fast removal of the voltage, the HV circuit breaker must first open before the TSC/TCR/TSR disconnector can be opened. The SVC shall be re-connected (HV breaker closed) after the TCR/TSR/TSC disconnector has been opened. The SVC shall auto reclose (HV breaker closing) after the TCR/TSR/TSC disconnector has been opened or filter circuit breaker/load break switch has opened. The SVC shall be designed to safely auto reclose regardless of the operating point prior to the tripping. The operating point shall be in accordance with V/I characteristics as specified in the SOW/TS. The SVC shall not auto reclose for busbar or main transformer failures.

7.3.3 For TSC, TCR/TSR faults such as the loss of redundancy of the thyristors, and cooling system problems the thyristors can be blocked. The HV circuit breaker must open first before the TCR/TSR/TSC disconnector can be opened. The SVC can be reconnected (HV breaker closed) after the TSC/TCR/TSR disconnector has been opened.

7.3.4 After any failure has occurred on the filter branch or the detuned capacitor bank the HV circuit breaker must be first opened before the relevant filter/capacitor bank load break switch or in case of MV circuit breaker and disconnect switch option the HV circuit breaker must be first open before the MV circuit breakers and disconnect switches open. The SVC can be re-connected (HV breaker closed) after the filter/capacitor bank load break switch, or MV circuit breakers and disconnect switches have been opened.

7.3.5 For auxiliary transformer faults, the relevant MV breaker shall be tripped where after disconnectors are opened without opening the HV breaker.

## 8.0 MANUAL CONTROL – FIXED ADMITANCE OPERATING MODE

The control system shall be provided with facilities for manual current control, which shall enable setting of the reactive output by direct control of the thyristor firing angle i.e., a fixed admittance operating mode, with the automatic control system out of service. The manual control operation shall only be allowed within the limits of the

SVC rating and its components and shall not cause automatic under- or overvoltage conditions on the network. Under- and overvoltage limits shall be selectable.

The manual control shall :

8.1 be provided with a follow-up device or similar system such that a transfer from automatic control to manual control mode and vice versa at any operating point within the SVC range results in a reactive disturbance of less than ± 5 MVAr. It shall not provoke voltage kicks or surges. This follow-up device would be disabled when the SVC is operating under manual control. The rate of adjustment during transfer from manual to automatic control mode shall be approximately 5 MVAr per second.

8.2 be capable of adjusting the unit output over its complete range.

8.3 permit testing of the control system with the SVC connected to the SEC/National Grid SA transmission system.

8.4 The activation and deactivation of this mode as well as the admittance reference must be controllable by commissioning/engineering personnel only.

8.5 Indicating instruments and appropriate controls shall only be accessible via password controlled access to the Human Machine Interface (HMI) that shall facilitate the transfer between automatic and manual mode and reference setting. Only remote indication of the control mode (automatic and manual) shall be required. This mode shall be automatically disabled and the controller is switched to the normal voltage control mode within 10 ms of detecting a system undervoltage or overvoltage condition.

8.6 All control system protective functions such as current and voltage limitations shall remain in operation while operating in manual mode.

## 9.0 AUTOMATIC ENERGIZING – SEQUENCE CONTROL

Automatic energizing of the SVC shall be provided and shall cause minimum disturbance only on the network.

9.1 Trip and Lock-out shall only occur for serious faults. Loss of AC Auxiliary power shall not prevent a restart. Loss of HV AC and based on a low battery level alarm shall result in the SVC switching off (#2 OFF) without any lock-out and restart shall be possible.

9.2 No alarms shall prevent restart after a normal shutdown: OFF

9.3 Manufacturer shall design the automatic sequence control taking into consideration that SVC connection to HV bus arrangement (breaker and half, double bus etc.). The manufacturer shall develop the "ON", and "OFF" sequence according to best international practice for such an arrangement. Manufacturer shall take note of the location of HV voltage measuring point.

9.4 During energizing of the SVC (ON), the SVC shall not cause a transient change in the voltage. The limit of the transient voltage variation with reference to the existing voltage is 5%. The manufacturer shall specify the guaranteed transient

voltage variation and duration as well as the energization time, including transformer inrush, etc. The manufacturer shall describe the sequence of energizing of the SVC (SVC transformer, TCR release, TSC released, FC energized etc.).

9.5 Within the control and protection system, facilities shall be included for the local and remote initiation of each of the following three functions: on, off and trip. Sequence control plant and material shall be provided to cause these initiations to perform the following functions:

9.5.1 #1 ON

The voltage reference shall be adjusted by the sequence control plant and material such, when the SVC is automatically connected to the power system, the output of the SVC does not exceed 5 MVAr capacitive or inductive. Following the energizing of the SVC by the closing of the SVC HV circuit breaker, and after a suitable time interval, the voltage reference shall be adjusted to the set point, the rate of adjustment being such that the reactive output ramps at 5 MVAr per second. If the system voltage exceeds an overvoltage threshold, the MVAr adjustment shall be ignored and the rapid voltage SVC controller enabled without any delay. In this case, the energizing of the SVC shall not last longer than 200 ms, after which the SVC voltage controller shall be active.

The SVC shall control the filter branches/capacitor bank by selective energization of each branch after the SVC has been switched ON.

9.5.2 #2 OFF

The voltage reference shall be adjusted by the sequence control plant and material to cause the reactive output of the SVC to become zero and the SVC subsequently disconnected from the power system by the opening of the SVC HV circuit breaker. The rate of adjustment shall be such that the reactive output ramps at a rate of approximately 5 MVAr per second. If the system voltage should drop below or exceed the adjustable voltage threshold levels (1.00 p.u. ≤ Umax ≤ 1.10 p.u. and 0.95 p.u. ≤ Umin ≤ 1.00 p.u. (step size of 0.01 p.u.)), the OFF program shall be stopped and the SVC Controller shall continue as before. Local and remote indication of this condition shall be provided.

9.5.3 #3 TRIP

a. The unit shall immediately be disconnected from the system with no delay to initially zero the reactive output. The thyristor valves shall be set to reduce the reactive output to zero simultaneously with the trip output signal to the SVC HV breaker.

b. A trip signal shall always have an immediate effect and shall

override both the ON and OFF functions.

c. This signal must be wired into the protection and control systems from the main substation control room and a local trip switch.

## 10.0 SLOW REGULATION OF REACTIVE POWER OR CONSTANT MVAr CONTROL

The SVC shall have a selectable "auto runback" option where the SVC shall reduce (ramp) its MVAr output to a pre-set operating point after an adjustable pre-set time delay dependent upon required SVC output.

10.1 This automatic device is designed to restrict the controlled reactive power range in the steady state condition. It shall ensure a sufficient reserve of reactive power for voltage control during major disturbances.

10.2 This device regulates the operating point (Q:MVAr) of the SVC to a value defined by a reference that can be continuously adjusted over the entire controlled SVC range. The setting must be within 1% of the rated current or SVC MVA rating (step size tolerance of 1 MVAr and step size of 1 to 5 MVAr). The reference can also be adjusted by the operator. This adjustment shall be from local and remote (raise and lower signals).

10.3 The Constant MVAr controller shall operate slowly so as not to interfere with the rapid voltage regulation and shall not cause under- or overvoltage conditions. The response time of the slow SVC controller shall be continuously adjustable from 1.0 to 600.0 seconds (step size of 1 second).

10.4 The SVC shall return to a voltage control mode if the busbar voltage drops below a pre-set value or exceeds a pre-set value. The output of the slow reactive power SVC controller is limited by independent positive and negative ceilings that are continuously adjustable between 0 and 0.1 p.u. (step size of 0.01 p.u.) of the reference voltage. These limits shall be of the dynamic type (non wind-up), i.e., once the output of the slow SVC controller has peaked, it must drop off as soon as the input signal changes polarity. The reference voltage signal (Vref) shall be reset to value given on HMI.

10.5 A voltage disturbance prompts (initiates) one of two actions:

10.5.1 The output of the slow reactive power SVC controller is reset and maintained at zero,

10.5.2 The output of the slow reactive power SVC controller is maintained at the value prior to the voltage disturbances.

A voltage disturbance is detected upon the occurrence of an undervoltage below an adjustable threshold of 0.9 to 1.0 p.u (step size of 0.01 p.u.) or an overvoltage above an adjustable threshold of 1.0 p.u to 1.1p.u. (step size of 0.01 p.u.).

The choice of action and adjustment of the voltage thresholds shall be selected by authorized personnel. Drop-out of the slow SVC controller during a voltage disturbance must be indicated (local and remote).

10.6 The effect of either actions described above is maintained for an adjustable period of 1 sec. to 50 min. In the event of the output of the slow SVC controller being held at the value prior to the voltage disturbance, the drift of the output value must be negligible throughout that period. Operator re-activation of the slow SVC controller must be possible, even if the timer-controlled period has not fully elapsed.

10.7 Manual deactivation of the slow SVC controller shall not cause voltage steps. Nor shall such voltage steps occur upon manual activation or manual or automatic re-activation of the slow regulation.

10.8 This automatic device must be compatible with the shared control of the SVCs at the substation, or generators/SVC's/HVDC at any substation electrically close.

10.9 The SVC controller shall allow enabling or disabling of this function from local or remote, including status indication and raise and lower of this signal from local and remote.

10.10 In case this reactive power control function is disabled and the measured reactive current output of the SVC remains within adjustable capacitive and inductive limits for adjustable periods, a status alarm indicator is made locally and remote.

## 11.0 CONTROL OF SHUNT DEVICES

### 11.1 General

The SVC shall automatically control the switching of external shunt capacitor banks and reactors which are connected at the MV Bus/Substation. The function shall be selectable.

The SVC shall be required to have provision (minimum six (6) external devices) for controlling the switching of shunt capacitors and reactors at the Substation. The control of external devices shall be co-ordinated with all other control functions as required in this standard to ensure stable and correct operation. The maximum time delay between the signal processing and output (to the Substation Control System Interface) shall be less than 10ms.

11.1.1 A signal to switch an external and internal capacitor or reactor shall be provided via a dedicated I/O system connected at the same and/or other busbars or voltages. The switching logic shall use the SVC admittance (Y) or thyristor firing angle as the evaluation signal. When this signal reaches predetermined set-points the switching sequence must be initiated with predetermined time delays. Both the set-points and time delay intervals shall be adjustable over a range of values with the input value between 0 and 1 p.u. (step size of 0.1 p.u) and time delay between 0.00 and 60.00 seconds (step size of 10ms). The program cycle time shall be less than 10ms.

11.1.2 The external device control logic must be co-ordinated with the Q-controller and individual device protection functions.

11.1.3 The control of external devices shall include a suitable hysteresis in

the switching control to avoid hunting.

11.1.4 The SVC controller shall allow enabling or disabling of this function from local or remote, including the status indication.

11.2 Design Criteria

11.2.1 The External Device Switching Logic (EDSL) shall assist the SVC's to optimize their dynamic output range. When a SVC reaches the end of its range either due to fast transient or slow load change, an external device shall be switched to bring the SVC back to its preferred dynamic operating range.

11.2.2 The External Device Switching shall be fast enough to reduce voltage collapse of the network.

11.2.3 The set points of the External Device Switching Logic shall be coordinated with the reference of the Q-Control to prevent hunting and excessive switching of the external devices.

11.2.4 The External Device Switching Logic shall operate automatically to assist the network operator in maintaining a stable network voltage without excessive switching of the external devices.

11.2.5 A capacitor bank can be switched IN and OUT without restriction, but it can only be switched OUT and back IN after a settable discharge time. This shall be considered when sequence of switching devices is derived.

11.2.6 The EDSL shall not duplicate or be in conflict with the existing protection system.

11.2.7 Implement discrepancy functionality for breaker and link status. If a switchable device is not OK due to discrepancy or communication failure, the specific device becomes not available and sends an alarm to the Power Control Centre (PCC). If a bus coupler or bus section is not OK due to discrepancy or communication failure, the complex EDSL becomes not available and sends an alarm to PCC.

11.2.8 All input signals shall be debounced.

11.2.9 The EDSL control the automatic / manual operation of the switchable devices. This is not available for the SVC capacitor banks.

11.2.10 If a device is not reacting to switching instruction the device shall become unavailable set to manual operation and an alarm shall be sent to PCC.

11.2.11 When a device is switched a signal must be sent to RTU/SAS to indicate which device was switched by EDSL.

11.2.12 If SVC is not available, all devices shall be switched to manual.

11.3 Operation

11.3.1 Manufacturer shall determine which devices are electrically connected to SVC.

11.3.2 Manufacturer shall derive the connection logic to determine the electrical connectivity.

11.3.3 The manufacturer shall determine the switching sequence of shunt devices.

11.3.4 Adjust the switching point according to the next device that shall be switched.

11.3.5 Use the TCR output to determine when a switching operation is required.

11.3.6 Switching Sequence

When a SVC reaches its limit, devices that are electrically connected to the SVC are switched according to a predetermined order:

When the EDSL is initialized the devices are prioritized and ordered according to their device numbers in each group. Once a device is required to be switched, the availability and status of the device are considered. If device on the list is not available, the next available device shall be switched. Each time a device is switched, the switching priority of all the devices are updated for that group by rotating the devices and placing the device that was switched at the back of the list. By rotating the devices in a group facilitate an equal number of breaker operations.

11.4 Positive And Negative Sequence Control

Following is tentative design for the positive and negative sequence control. The Manufacturer/Contractor may suggest his own standard design within the boundaries specified SOW/TS:

11.4.1 Closed loop voltage control:

The voltage at the SVC HV bus in the substation shall be measured.

The measured voltage shall be separated in phase sequence components and DQ transformed. A negative phase sequence current counteracting the measured negative phase sequence voltage shall be generated in the SVC. Separate PI (proportional - integrating) regulators shall be used for the D and Q components of the measured negative phase sequence voltage to determine the needed phase wise different SVC susceptance.

A positive phase sequence current giving the wanted positive phase sequence voltage shall be generated in the SVC. A PI (proportional -integrating) regulator shall be used for the Q component of the measured positive phase sequence voltage to determine the needed

phase wise identical SVC susceptance. The phase wise calculated susceptances shall be summed together.

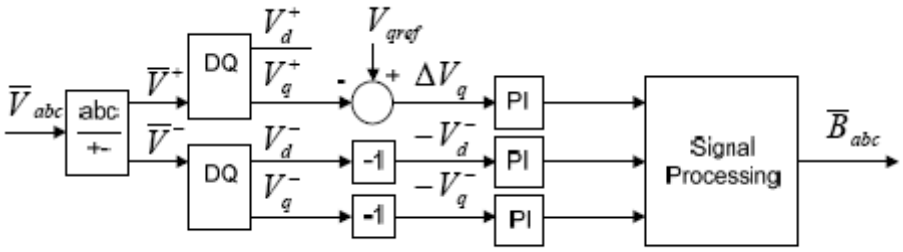



Figure: Positive and Negative Sequence Control

Each one of the control functions described above i.e, closed loop negative phase sequence voltage control and positive phase sequence voltage control shall be made available for enabling/disabling on the HMI. The controller shall be able to run with any combination of the control functions enabled.

Unless otherwise specified in SOW/TS the SVC shall be able to use one half of its full power for unbalanced voltage control. The range of the SVC that can be used for unsymmetrical control shall be limited only by the available TCR/TSR and TSC range.

The SVC shall be able to run phase wise unsymmetrically during faults and directly post fault recovery. For each one of the TSRs and TSCs, one, two or all three phases may run in order to give coarse unbalance control. The fine unbalance control is performed by the TCR.

The SVC shall fully balance the negative phase sequence voltage (if within SVC power capability), or bring it below a preset value (dead-band set from HMI), at its connection to the grid.

Optimal strategies shall be developed to determine the controller operation when one or several of the SVC delta/wye phases hit their maxima/minima.

The controller shall remain stable for transient and other dynamic operation of the network and the SVC.

Unless otherwise specified in SOW/TS the SVC shall be designed for unsymmetrical operation of minimum one half of its full range during system faults and post-fault recovery. Unbalanced operation of TCR/TSR/TSC is required. Control strategy during power system short circuits shall be determined and verified by studies.
SEC/National Grid SA aims for maximum voltage support for induction motor loads from the SVC. Unsymmetrical voltage control shall be used in order not to violate voltage limits (line to

line voltages or line to ground voltages) at PCC during and after grid short circuits.

## 12.0 SELF-CHECKING OF AVAILABILITY OF COMPONENTS

The SVC control system shall automatically self-check the availability of all components. Self-checking of power electronic valve branches shall not violate harmonic distortion limits nor shall there be changes in the SVC output. There shall be provision for the operator to manually initiate self-checking.

Thyristor valve branches, TCR(s) and TSC(s), shall be operated using permutation sequence. There shall be no restriction in the master-follower sequence. All TCRs shall be designed to operate both as TCR and TSR without violating the requirements stated in this standard.

## 13.0 SHARED CONTROL OR PARALLEL OPERATION

The SVC shall be capable of operating in parallel with another SVC or generators without reducing or influencing the efficiency or range of the SVC during any of the above mentioned functions. The parallel SVC may be located at the same substation or at any substation electrically close. The parallel SVC may even be at a different voltage. Refer to Scope of Work/Technical Specification (SOW/TS).

The SVC response time defined above shall be achieved with the SVCs operating alone or in parallel with one another.

## 14.0 SUPRESSION OF HARMONICS AND FILTER PERFORMANCE

### 14.1 Harmonic Performance Limits

14.1.1 The maximum voltage distortion generated by the SVC shall not exceed the maximum voltage harmonics limits. The harmonic limits shall be met for symmetrical as well as for unsymmetrical SVC operation. The limits of the individual harmonic corresponding to the highest background harmonic value shall be the remaining value after subtracting the ambient harmonics arithmetically from the limits stated in the Saudi Arabian Grid Code (SAGC)**.** The limits (of each individual harmonic as well as the THD) shall be 70% of that remaining after the ambient harmonics except for the highest ambient harmonic value (as noted in SOW/TS) is subtracted geometrically from the limits stated in the Saudi Arabian Grid Code (SAGC) . This shall be limited to harmonic orders as provided in the SOW/TS.

For the highest individual background harmonics (VBACKGROUND, MAX), the following formula shall be applied V*Limit=SAGC LIMIT-V BACKGROUND*

For the rest of each individual background harmonic voltage (VBACKGROUND), for the calculations shall be considered below.

$$VLIMIT=(\sqrt{SAGC^2LIMIT- V^2BACKGROUND} ) \times 0.7$$

14.1.2 The maximum current distortion generated by the SVC shall not cause the values shown in Table 3 & 4 of IEEE Std 519-2014 to be exceeded.

14.1.3 These requirements should be met for the continuous range of system and environmental conditions stated in this specification and the parameters detailed in this section such as:

a. Continuous range of the SVC

b. Variation in tolerance of total filter capacitance, including permissible fuse failures.

c. Variation in tolerance for SVC parameters, such as transformer winding unbalances, system unbalance, valve firing variations, and unequal reactor and capacitor reactance between phases.

14.1.4 The applicable frequency range for the harmonic limits are stated in SOW/TS.

14.1.5 All harmonic performance and rating calculations shall be performed with a capacitance variation due to temperature that can be supported by tests. The conditions for the calculations shall be energized capacitors at a minimum ambient temperature to a maximum ambient temperature as specified in latest revision of 01-TMSS-01. If no relevant tests are available, new test have to be carried out before approval of filter design.

14.1.6 In case of one filter branch out of service (degraded mode operation) the harmonic voltage distortion limits (VLlMIT) are increased by 100% but they shall not exceed the limits stipulated in the Saudi Arabian Grid Code (SAGC).

14.1.7 The harmonic requirements (limits) shall apply to the SVC configuration with all filter branch group in service.

14.1.8 The ratings of all the equipment shall be designed to withstand continuously the distortion during all permissible degraded modes such as one filter branch out, etc.

14.1.9 These performance requirements may be exceeded for system conditions outside the continuous normal envelope as specified in the SOW/TS.

For rating purposes, the SAGC limits shall be used as background harmonic source voltage; the individual harmonic contributions shall be limited such that the specified THD limit is not exceeded. This is done such that a different set of individual harmonics may be selected for each DRPC component, depending on which harmonics give the highest stresses for that particular component.

As a first step the raw calculated harmonic stresses for the given

component are sorted in descending order, then using this order, harmonics stresses are added up starting with the largest harmonic contribution and adding successively smaller harmonic contributions, also adding the individual source voltages until the chosen THD limit for the harmonic source is reached. The following limits on the maximum contribution limits for individual groups of harmonics shall also be respected while performing these additions, as follows:

a. Contribution of 3rd, 5th and 7th harmonic is max. 70% of THD limit.

b. Contribution of harmonics >15 is max. 50% of THD limit.

The above individual harmonic contribution limits in a & b shall be waived for specific harmonic order if the value of the background harmonic contribution for a specific harmonic order is exceeding this limit already and hence the measured values for that harmonic order shall be used.

c. THD of the corresponding source voltage is limited to the specified THD limit.

Limiting the total THD of the corresponding source voltage to the specified THD limit cuts off the remaining harmonic contributions and reduce them Proportionally, conditioned that the remaining harmonic values shall not be below the background harmonic values. i.e., all subsequent harmonic contributions are then used equal to the background harmonic values.

d. Manufacturer/Contractor shall perform comprehensive background harmonic measurements for each SVC location. The harmonic measurements shall be carried out as per following requirements;

i. The assessment period shall be at least 10 days.

ii. Measurements shall utilize wound (electromagnetic) voltage transformers. The best available existing VT in the substation shall be used for the measurements. SEC/National Grid SA will define the VT to use.

iii. Equipment: The instrument accuracy shall be (minimum) class B, as specified in IEC 61000-4-7.

iv. Measurement interval: Measurements shall be done in accordance with IEC 61000-4-7. The recording interval shall be 10 minutes while multiple points shall be sampled over the interval period, but only the 95% value for each interval shall be saved together with Min, Average and Max values.

v. Following power quality quantities shall be recorded; Unbalance, Individual harmonics (voltage and current), THD

vi. Data Extraction: The data shall include the Min, Average, Max and 95% values.

vii. Voltage of bus bar and current flow in line feeders, transformer feeders, bus sections and SVC feeder shall be measured.

viii. Phase connection: The measurement probe shall be connected in star to obtain individual phase to ground voltage measurements.

ix. Current flow convention: The measurement equipment shall be connected that the arrow on all current probes shall point away from the bus bar for all feeders. This will ensure that the same convention be followed for all feeders.

x. Three phase measurement: The measurement for current harmonics shall be on all three phases for all measurement locations.

xi. Manufacturer/Contractor shall provide measurement procedure for SEC/National Grid SA approval.

xii. Results from measurements shall be presented in a technical report. Measurement data (results from measurement) shall be provided on CD/DVD in EXCEL format. Measured values shall be compared with values given in the SOW/TS.

xiii. Measurements shall be performed early during the base design stage.

14.2 Filter Design Requirements

14.2.1 Harmonic filters shall be designed in such a way that the total harmonic currents from the power system and the SVC do not overload the filter reactors, resistors and capacitors. The rated current of the filter components (including load break switches and circuit breaker) shall be based upon the calculation of the generated harmonic currents added quadratically (RSS) and adding 20% margin in total rated current (including fundamental and harmonic loading) in consideration of background harmonics.

14.2.2 The rated voltage of capacitors shall not be less than the arithmetic sum of the normal continuous power-frequency voltage and each of the individual harmonic voltages (from n=2 to 50) and an added margin of minimum 20% in total rated voltage (including fundamental and harmonic loading) in consideration of background harmonics.

14.2.3 The components of characteristic and non-characteristic harmonic filters shall be sized upon the following factors:

a. Absorption of characteristic and non-characterstic harmonic currents generated by the proposed SVC.

b. Filter detuning and maximum expected frequency deviation

which shall not cause harmonic overvoltages beyond a certain limit.

c. Requirements and contingencies described herein.

d. Operation of SVC with all redundant filters out of service.

e. Maximum current amplification

14.2.4 De-tuned capacitor banks shall be rated as per 14.2.1, 14.2.2 and 14.2.3 above. The fundamental frequency current rating shall be derived from the point giving highest continuous voltage on the medium voltage bus in the VI diagram.

14.3 TSC Design Requirements

14.3.1 TSCs shall be designed in such a way that the fundamental frequency current and the total harmonic currents from the power system and the SVC do not overload the reactors, capacitors and thyristor valves.

14.3.2 In case the calculations as per (14.3.4) and (14.3.5) results in higher current and voltage ratings than the method (14.3.3) with a voltage source behind the network, then these higher ratings shall be applied.

14.3.3 The harmonic rating of the reactors and capacitors from ambient harmonics shall be calculated using a harmonic voltage source behind the network impedance. The fundamental frequency current and voltage ratings shall be based on the worst operating condition including component tolerances.

14.3.4 The rated current of the TSC reactors, capacitors and disconnect switch shall be based upon the calculation of the fundamental frequency current and the SVC generated harmonic currents added quadratically (RSS), and adding a 10% margin in total rated current (including fundamental, and SVC generated harmonic loading) to consider background harmonics and other unknowns.

14.3.5 The rated voltage of capacitors should be not less than the arithmetic sum of the normal continuous power-frequency voltage and each of the individual harmonic voltages (from n =2 to 50), and an added margin of a minimum of 10% in total rated voltage (including fundamental, and SVC generated harmonic loading) to consider background harmonics and other unknowns, if the background harmonic loading does not exceed this margin already.

14.3.6 The components of TSCs shall be sized upon the following factors:

a. TSC and filter detuning and maximum expected frequency deviation.

b. Requirements and contingencies described herein.

c. Operation of SVC in all different degraded modes.

d. maximum current amplification

14.4 SVC Power Transformer Design

The voltage and current stress from ambient harmonics shall be calculated by using a harmonic voltage source behind the network impedance. The transformer current rating shall be the geometric sum of the maximum fundamental current, the background harmonic currents and the generated harmonics.

14.5 Harmonic Current Amplifications

Second harmonic resonance (large current & voltage amplification) seen from the SVC medium voltage bus shall be avoided to the maximum extent practical. Power transformer impedance or TSC tuning shall be optimized for low second harmonic amplification. Studies to be performed during base design shall demonstrate low second harmonic current or voltage amplifications. Sensitivity of TSC tuning and power transformer impedance shall be investigated. TSC tuning and/or power transformer impedance shall be changed if high second amplification exists. A amplification factor below 20 is acceptable.

Total current amplification for each harmonic shall be <20. Current amplification for individual branches, harmonic filters, de-tuned capacitor banks and TSC, shall be <10 for harmonics other than 2nd . Higher amplifications than the given limits above, by no more than 50%, can be compensated by increasing the rating corresponding to the specific harmonic order amplification increase by the same ratio.

14.6 Power Oscillation Damping

The commonly used lead–lag structure shown in below figure shall be chosen for damping controller. The lead-lag structure is preferred by the power system utilities because of the ease of on-line tuning and also lack of assurance of the stability by some adaptive or variable structure techniques. The structure consists of a delay block, a gain block with gain Ks a signal washout block and two-stage phase compensation block. The time delay introduced due to delay block depends on the type of input signal. For local input signals only the sensor time constants is considered and for remote signals both sensor time constant and the signal transmission delays are included. The signal washout block serves as a high-pass filter, with the time constant TW, high enough to allow signals associated with oscillations in input signal to pass unchanged. From the viewpoint of the washout function, the value of TW is not critical and may be in the range of 1 to 20 seconds. The phase compensation blocks (time constants T1S, T2S and T3S, T4S) provide the appropriate phase-lead characteristics to compensate for the phase lag between input and the output signals. Vref represents the reference voltage as desired by the steady operation of the system. The desired value of reference voltage is obtained according to the change in the SVC reference ΔV SVC which is added to Vref to get the desired voltage reference VSVC_ref. All the time constants and gain parameters should be available at the HMI to set the values.

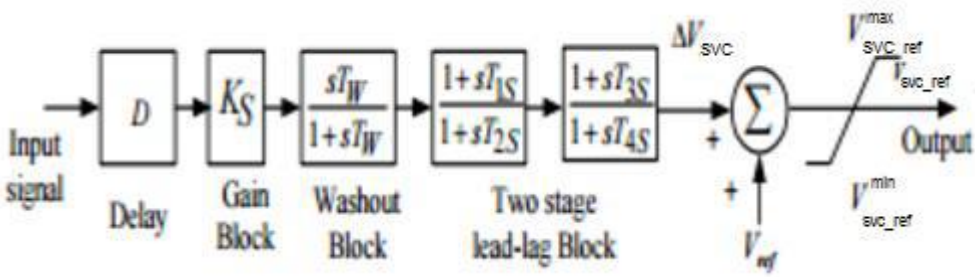



Figure: Power Oscillation Damping

## 15.0 TELEPHONE INTERFERENCE

The following limits shall not be exceeded:

15.1 Telephone influence factor (TIF) –35

15.2 Balanced IT - 25 000 weighted amperes

Refer to IEEE 519-2014 Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems.

## 16.0 SVC AVAILABILITY AND RELIABILITY

16.1 Definitions

SVC availability and reliability definitions shall be per clause 8.7 of IEEE 1031.

16.2 Required Availability and Reliability

16.2.1 The annual equivalent availability for forced outages for the SVC shall be at least 99.7%.

16.2.2 The maximum number of forced outages shall not exceed 2 outages per year.

16.2.3 Manufacturer shall submit guarantee average number and duration of scheduled outages per year.

16.2.4 Manufacturer shall guarantee SVC performance per this specification. If the actual performance of SVC is below the value, manufacturer shall provide corrections, modifications etc., to meet the guaranteed performance without any extra cost to SEC.

16.2.5 Manufacturer shall state maintenance interval and also shall state monitoring of the equipment.

## 17.0 MAGNETIC AND ELECTRIC FIELD STRENGTHS

The clearances shall be so that no person outside the fences area or any area accessible while the SVC is energized shall not be exposed to magnetic field strength of more than 0.5mT (5.0 Gauss) and electrical field strength not more than 10kV/m. Guidelines per IRPA shall be used.

## 18.0 OVERALL DESIGN CRITERIA

### 18.1 Plant and Material Characteristics

All plant and material shall be entirely suitable for continuous operation under the conditions specified without any restriction to the operating range of SVC. Plant and material shall include everything necessary for safe and conventional installation, operation and maintenance of the plant and material whether specified or not. SVC shall operate with minimal maintenance for minimum 25 years over its entire range under most severe conditions per this standard.

No components shall be overloaded or disconnected under any SVC operating condition particularly auxiliary AC voltage range and MV operating voltages and harmonics.

### 18.2 Layout and Design

SVC layout and design shall be modular with each branch section separated with fencing and having suitable access control (fence with lockable gate and interlocking). Sufficient space shall be provided between the plant and material and the fence to allow maintenance and repair access. Fire wall between transformer shall be provided per TES-P-119.21. Access road shall be provided. Unless otherwise stated in the SOW/TS transformer rail shall be per latest revision of 53-TMSS-01.

Layout shall be arranged in such a manner that access to and removability of any plant and material is possible without disconnecting other plant and material.

### 18.3 Fire Retardant Material

Fire retardant material shall be used in the SVC especially in the thyristor valves and other components.

### 18.4 Losses and Loss Evaluation

#### 18.4.1 Graph for Losses

Manufacturer shall submit a graph and a table for the total operating losses over the complete steady state operating range of the SVC at system voltage of 1p.u. The losses shown on the graph and table shall be based on calculation of the losses of each of the components of SVC according to the method outlined in IEEE 1031. Calculated & Guaranteed losses for each component shall be submitted. For transformer losses shall be per latest revision of 53-TMSS-01. Losses shall be calculated with maximum ambient temperature of 55ºC & annual average ambient temperature of 35ºC for outdoor per 01-TMSS-01, 1.0 p.u on the HV bus and slope setting of 0%.

Losses in the table shall be shown in steps of 5 MVAr SVC output.

18.4.2 Cost of Losses and Loss Evaluation

Shall be as specified in SOW/TS.

18.4.3 Penalties when Losses Exceed Guaranteed Value

For transformer, guaranteed losses value shall be as specified in the latest revision of 53-TMSS-01. Otherwise, penalty shall be imposed per SOW/TS/Contract or transformer shall be rejected per transformer standards. Similarly, SVC component guaranteed losses with tolerance shall not exceed when the losses are measured. Otherwise, Penalty as specified in the SOW/TS/Contract shall be imposed when losses exceed up to 5% above guaranteed losses with tolerance. Beyond 5% losses, the equipment shall be rejected. Loss evaluation shall be based on the formula specified in the SOW/TS.

## 19.0 SVC AUXILIARY SUPPLY

### 19.1 AC auxiliary Supply

Unless otherwise specified in the SOW/TS two (02) station auxiliary transformers shall be provided for SVC. AC auxiliary supply shall be per TES-P-119.30 with automatic transfer scheme. In addition RMU shall be provided for AC auxiliary supply for all substation (from 69kV through 380kV).

### 19.2 DC Auxiliary Supply

DC auxiliary system for SVC shall be provided with Two (02) No. station battery each rated 100% and Two (02) No. battery charger each rated 100%. DC distribution scheme shall be per TES-P-119.30.

## 20.0 SVC SHIELDING , GROUNDING AND LIGHTING SYSTEM

Shielding, Grounding and Lighting system for SVC shall be designed per TES-P-119.07, TES-P-119.10 and TES-P-119.25 respectively.

## 21.0 SVC SYSTEM STUDIES

System studies as specified in the SOW/TS shall be carried out.

## 22.0 THYRISTOR VALVE COOLING SYSTEM

Thyristor Valve Cooling System shall be as per 59-TMSS-02 “Valves Cooling System For Dynamic Reactive Power Compensator (DRPC)”.

## 23.0 NOISE LEVEL

Noise level at the substation fence shall be per clause 5.2.6 of TES-P-119.02. Unless

otherwise specified noise level of SVC transformer shall be per NEMA TR-1. However transformer noise level, and noise level from other equipment such as thyristor valves shall be reduced to achieve the noise level specified in TES-P-119.02 at substation fence. Necessary noise level measurement shall be carried out to prove the same.

## 24.0 THYRISTOR VALVES

Thyristor Valves shall be per 59-TMSS-01 "Thyristor Valves For Dynamic Reactive Power Compensator (DRPC)".

## 25.0 SVC EQUIPMENT

The following equipment shall be provided per SOW/TS:

25.1 SVC Control HMI

25.2 SVC Control and Protection, including measurement, monitoring and indication.

25.3 SOE (when SAS is not specified for SVC), Disturbance Fault Recorder and DSM

25.4 SVC Transformer, SVC auxiliary equipment etc.

## 26.0 BIBLIOGRAPHY

26.1 IEEE 1031, "Guide for the Functional Specification of Transmission Static Var Compensators", 2011.

26.2 Cigre Brochure #25, "STATIC VAR COMPENSATORS", 1986.

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